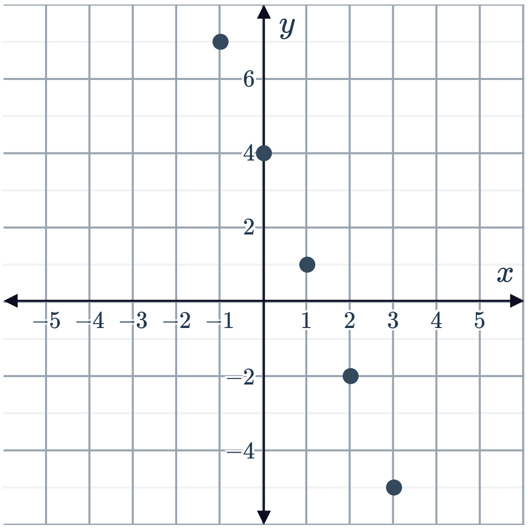
Appropriate models for data

1

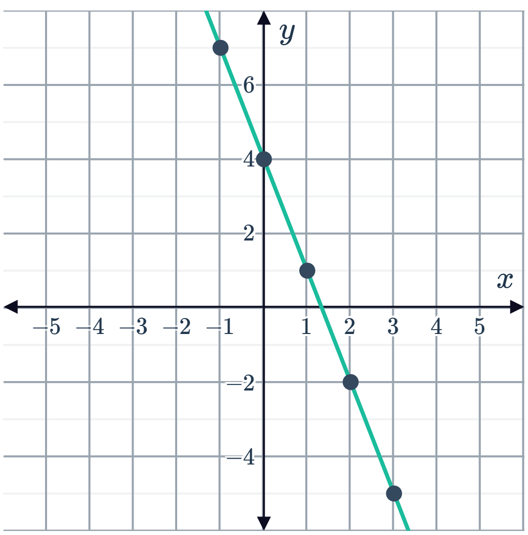
a

i



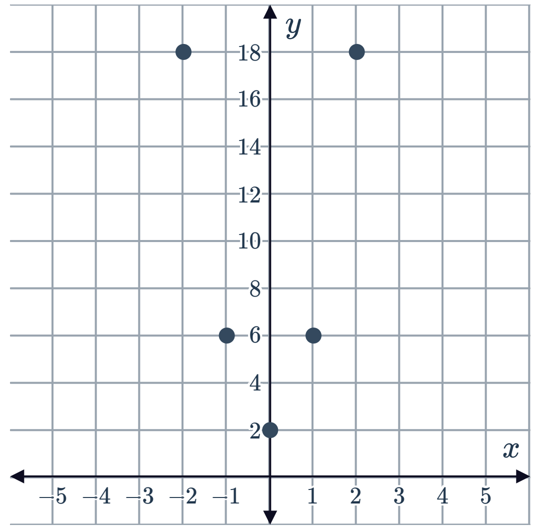
ii Linear

iii



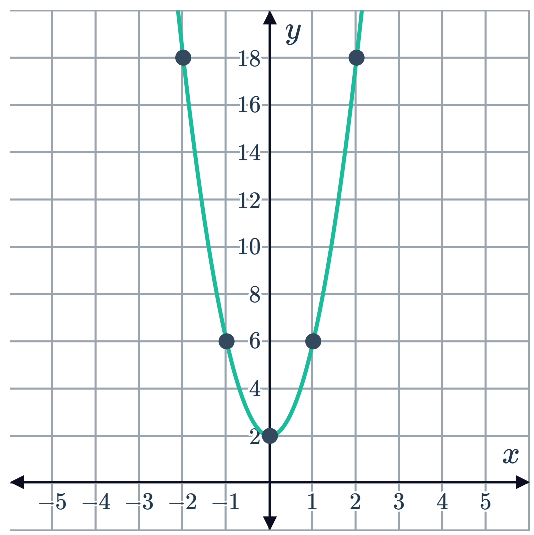
b

i



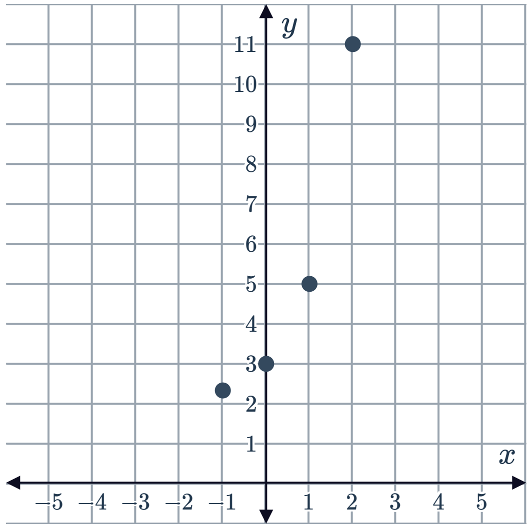
ii Quadratic

iii



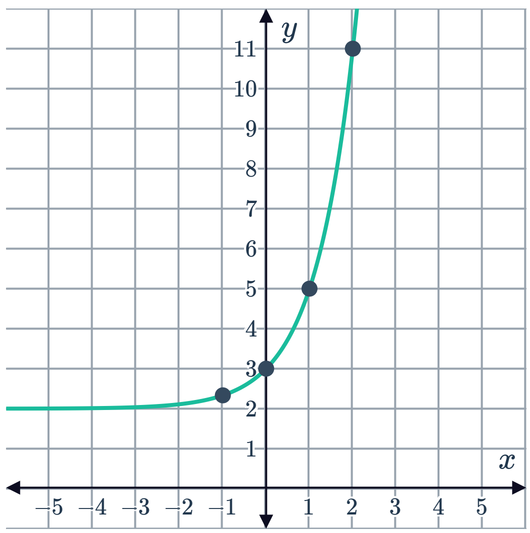
c

i

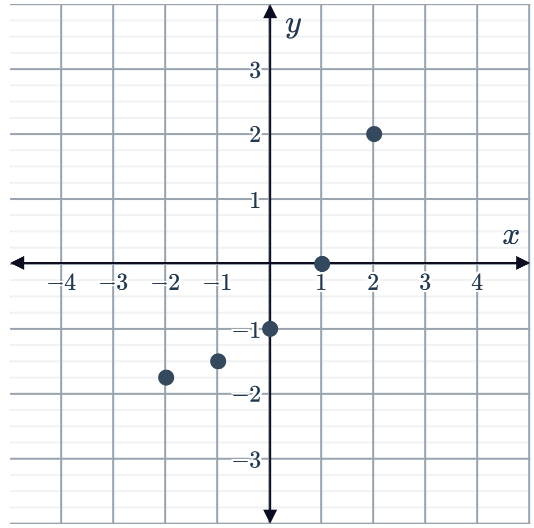


ii Exponential

iii

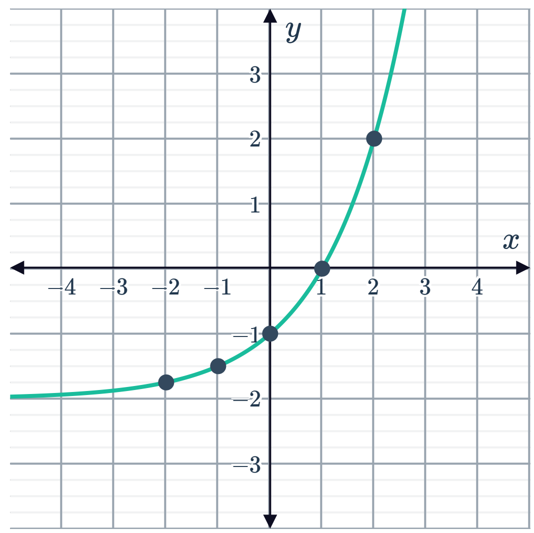


i



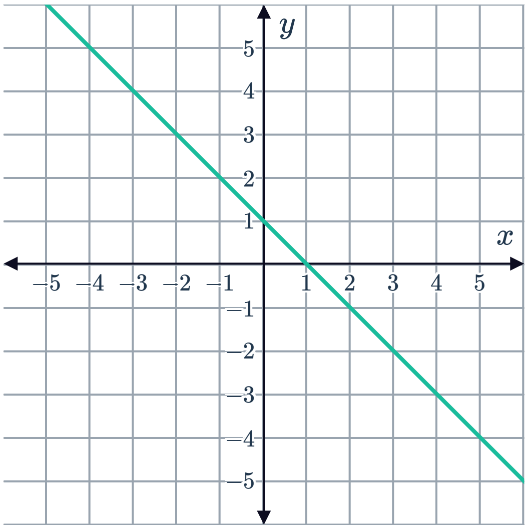
ii Exponential

iii



2

a



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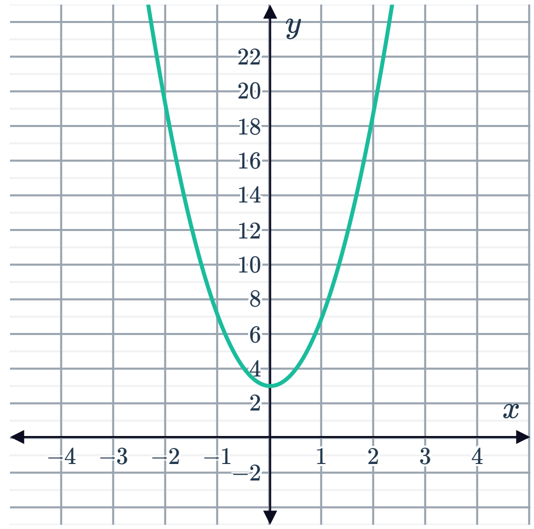

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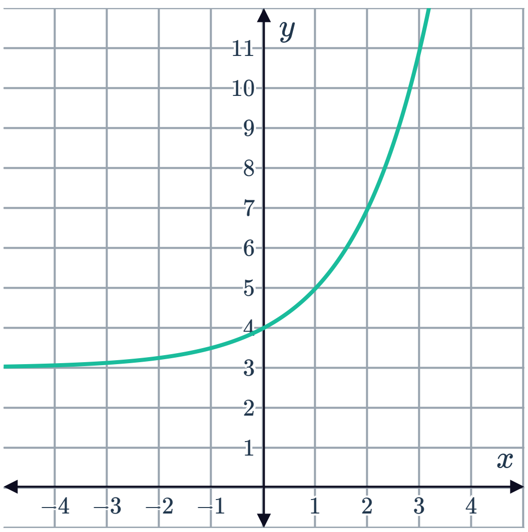
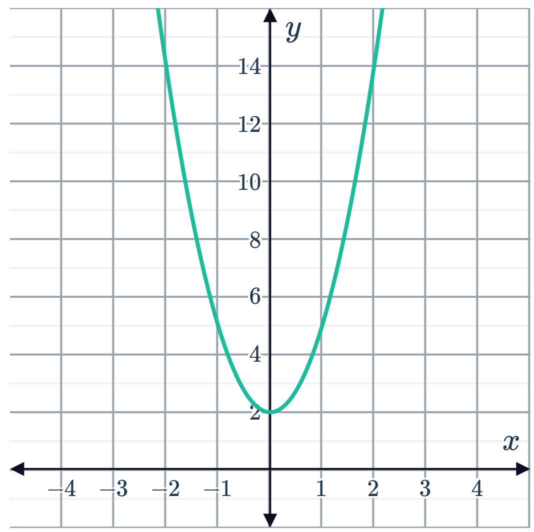


d



3

a A linear model would be the best way to model the data because the points are close to being along a straight line.

b A linear model would not be the best way to model the data because the points are close to being along a curve rather than along a straight line.

4

a Linear, $f(x) = mx + b$

b Quadratic, $f(x) = ax^2 + bx + c$, $a > 0$

c Quadratic, $f(x) = ax^2 + bx + c$, $a < 0$

Equations of models

5

a $y = a(x-h)^2 + k$

b $y = 2(x+3)^2 - 3$

6

a $y = a(x-p)(x-q)$

b $y = -(x-3)(x+1)$

7

a Cubic function

b i No ii Yes iii No iv Yes

c $2(x-1)(x-3)^2$

8

a Cubic

b i No ii Yes iii No iv No

c $y = -(x-1)(x-3)(x-4)$

9

a $y = a(x-h)^3 + k$

b $y = 2(x-1)^3 - 3$

10

a Rational

b i No ii No iii No iv Yes

c $y = \frac{2}{x} + 5$

11

a Rational

b i Yes ii No iii No iv No

c $y = -\frac{1}{x-4} + 3$

12

a $a = 5$

b $b = 2$

c $y = 5 \times 2^x$

13

a

i $c=3$ ii $a=3$ iii $b=2$

b $y=3\times 2^{-x}+3$

14

a $y=2\log_{10} x$ b $y=\log_{10}(x-2)$ c $y=\log_{10} 2x$

15

a

i Quadratic ii $y=\frac{x^2}{3}+6x+31$

b

i Rational ii $y=\frac{16}{x}$

Applications

16 Yes, for every week that passes, Rochelle's savings will increase by a constant amount.

17 Yes, the change in the volume of water in the bucket is constant over time.

18 No, for each unit of time, the change in height is not constant.

19 No, with every minute the change in temperature of the soup becomes smaller and smaller.

20

a

i No ii Yes iii No iv No

b 2

c 1 m

21

a 0

b No, there will ultimately be no Potassium-40 left in the sample.

22

a 0

23

a $24 \leq x \leq 30$

b $40 \leq x \leq 46$

c No - Jack's model is only accurate in the region $24 \leq x \leq 30$ and Georgia's model is only accurate in the region $40 \leq x \leq 46$, but there is a box used by the supermarket that has $x = 35$ which is in neither region.

24

a

i No

ii No

iii Yes

iv No

b No. The data points outside the region $3 < x < 7$ begin to diverge from the linear model.

25

a

i No

ii No

iii No

iv Yes

b The tree will stop growing and remain at a height of 700 cm.

26

a 5 seconds

b The model fits the data before the truck reaches the town.

c Yes

27 B

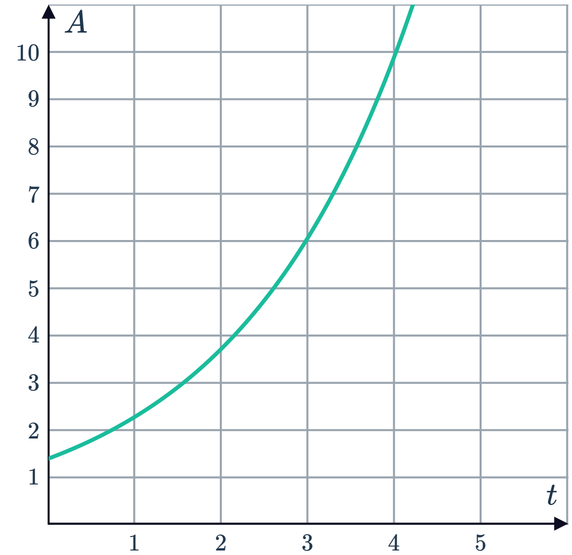
28 C

29

a

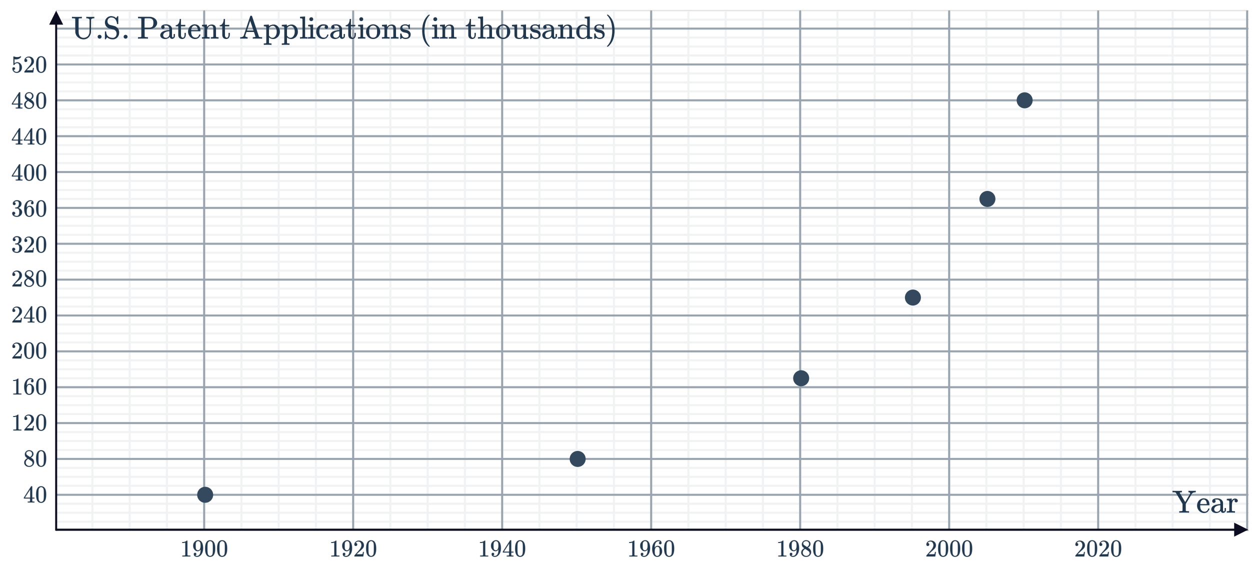
A	0	1	2	3
t	1.4	2.3	3.7	6.1

b



30

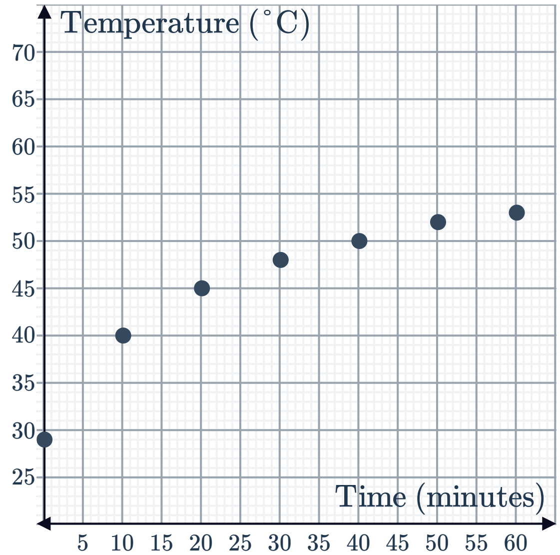
a



b Yes, an exponential model is appropriate because when drawn on a graph it has an exponential shape which, which is increasing more rapidly over time.
Also, if we calculate an exponential regression on this data set, we get $R^2 = 0.9627$ which is very close to 1, so there is a strong correlation to the exponential model suggesting an exponential association.

31

a



b A logarithmic function